Habit Shaper

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Engineering Coding Test Technical Project Planning & Architecture Specification

1. EXECUTIVE SUMMARY & CORE OBJECTIVES

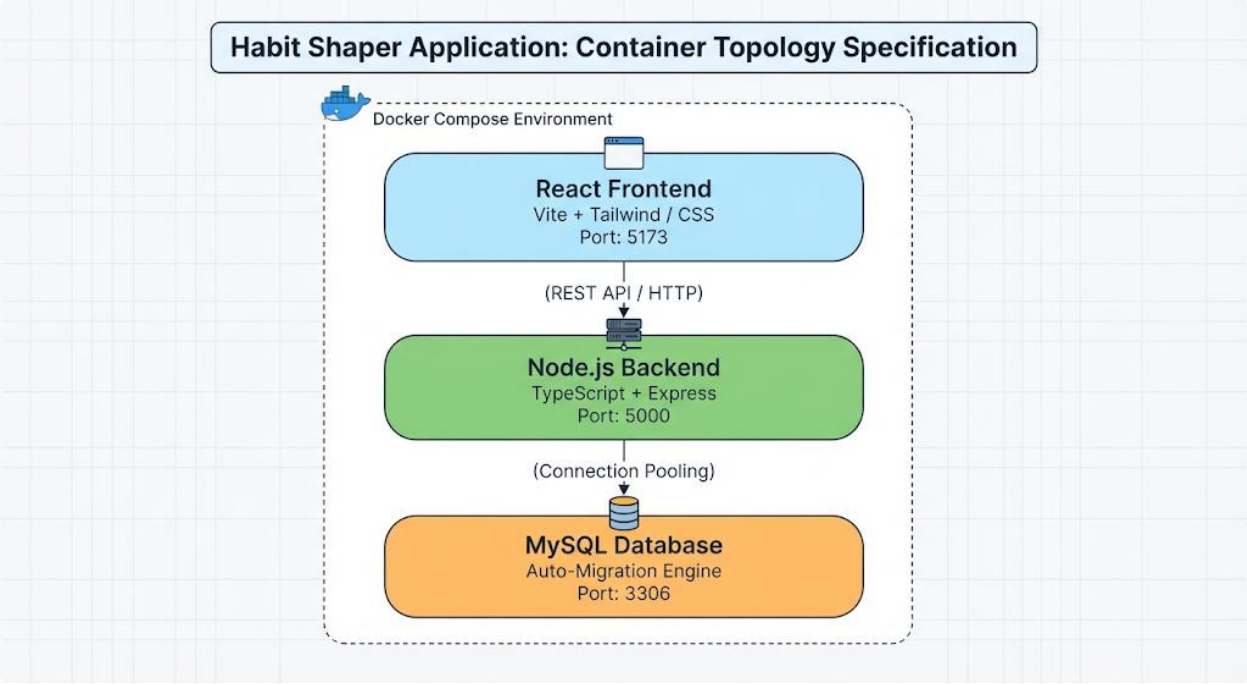
Habit Shaper is designed to provide comprehensive routine management through a modular, containerized architecture. The core functional objectives are split across three key pillars:

- **Habit Building:** Track positive routines, monitor daily completion, consecutive streaks, and analyze weekly success rates.
- **Habit Breaking:** Track clean streaks for negative habits, manage relapses safely, and re-establish clean milestones.
- **Goal Management:** Organize macro goals linked directly to specific build or break habits with full lifecycle controls.

2. SYSTEM ARCHITECTURE & CONTAINER TOPOLOGY

The application is fully containerized using Docker Compose, orchestrating three independent services interacting seamlessly:

- **React Frontend:** Vite + Tailwind/CSS | Port: 5173
- **Node.js Backend:** TypeScript + Express | Port: 5000
- **MySQL Database:** Auto-Migration Engine | Port: 3306



3. DATABASE SCHEMA DESIGN (MYSQL)

Relational mapping connects users to habits (1-to-Many), where each habit records daily logs in habit_logs and associates with milestone targets in goals.

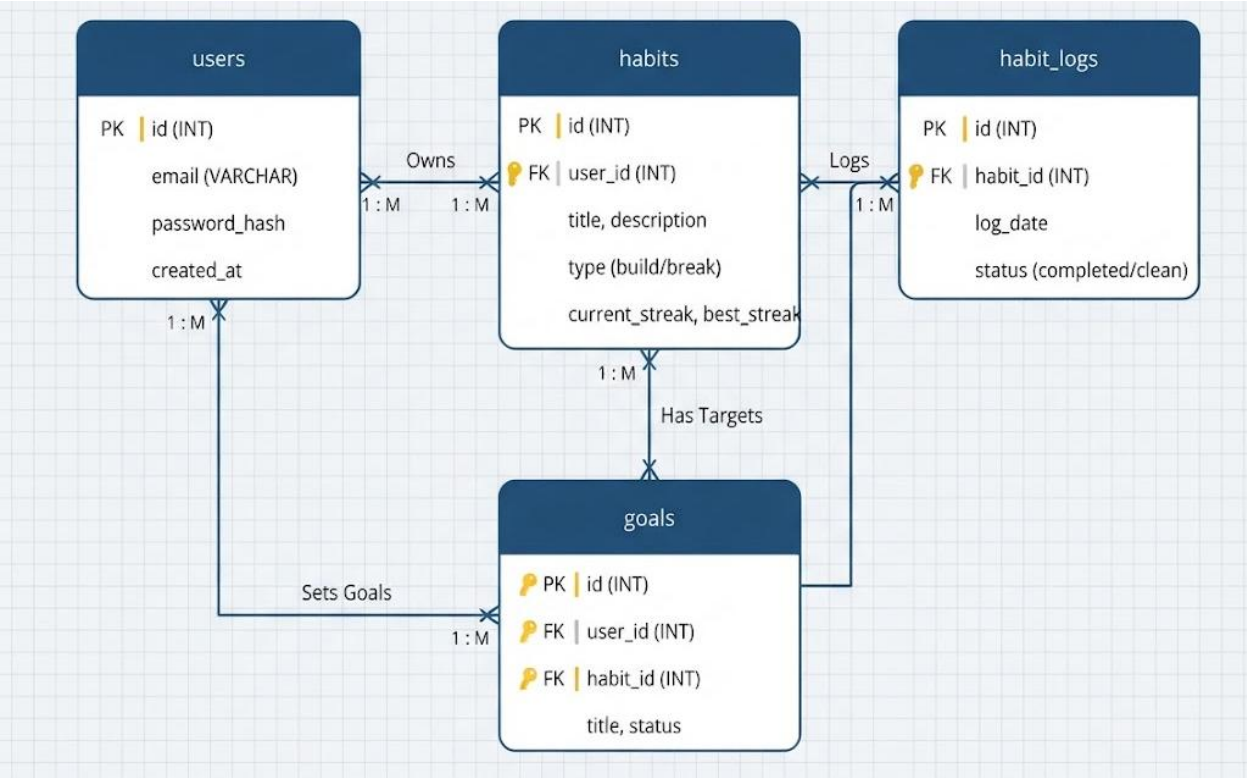


Table Name	Key Columns	Description & Constraints
users	id, email, password_hash, created_at	Stores user credentials with secure bcrypt password hashing.
habits	id, user_id, title, description, type, current_streak, best_streak, status	Defines habit goals categorized as 'build' or 'break' types.
habit_logs	id, habit_id, log_date, status	Records daily completion/clean status with unique composite keys.
goals	id, user_id, habit_id, title, status	Macro-level targets tied directly to specific active habits.

4. STEP-BY-STEP EXECUTION ROADMAP & CHECKLIST

Phase 1: Environment & Compose Setup

- ☐ Init git repository structure
- ☐ Create Dockerfile for backend
- ☐ Create Dockerfile for frontend
- ☐ Configure compose.yml orchestration linking MySQL, Backend, and Frontend containers
- ☐ Create .env.example & README with automated startup scripts

Phase 2: Backend Core & Database

- ☐ Setup Node.js TypeScript Express server
- ☐ Configure MySQL connection pooling
- ☐ Write auto-bootstrap migration SQL
- ☐ Implement User Authentication (JWT / Bcrypt)
- ☐ Setup route middleware protection

Phase 3: Habit & Goal Services

- ☐ Develop CRUD endpoints for Habits
- ☐ Implement Daily Log completion logic
- ☐ Build Streak calculator engine
- ☐ Develop CRUD endpoints for Goals
- ☐ Integrate Goal-to-Habit linking

Phase 4: Frontend UI & Integration

- ☐ Setup React + Vite + Tailwind
- ☐ Build Login/Register views
- ☐ Create Habit Tracker Dashboard
- ☐ Create Goal Management view
- ☐ Build API hooks for seamless state management

Phase 5: Final Polish & Verification

- ☐ Run full docker compose test
- ☐ Validate end-to-end containerized execution via single command docker compose up --build